

Лемма (св-ва мощности)

$$\exists S: \mathcal{U} \rightarrow \mathbb{R} \Rightarrow$$

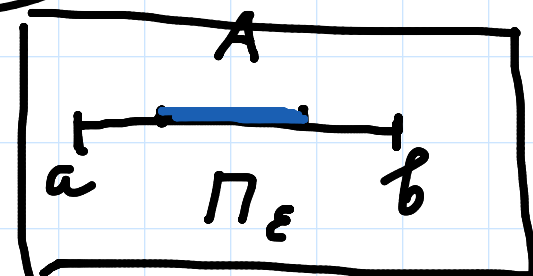
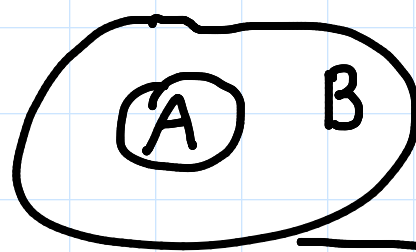
$$(1) \quad S - \text{монотонна: если } A, B \in \mathcal{U}, A \subset B \Rightarrow \underline{S(A) \leq S(B)}$$

$$(2) \quad \exists A \in \mathcal{U}, \exists [a, b]: A \subset [a, b] \Rightarrow \underline{S(A) = 0}$$

$$(3) \quad \exists A, B \in \mathcal{U}: A \cap B = \emptyset: S(C) = 0 \Rightarrow \underline{S(A \cup B) = S(A) + S(B)}$$

1) Т.к. $\begin{cases} B = A \cup (B \setminus A) \\ A \cap (B \setminus A) = \emptyset \end{cases} \xrightarrow{\text{def (2)}} S(A \cup (B \setminus A)) = S(A) + S(B \setminus A) \xrightarrow{\text{def (1)}} S(A \cup (B \setminus A)) > S(A) \Rightarrow S(B) > S(A)$ 3-й шаг

$$2) \quad \exists \varepsilon > 0 \Rightarrow \exists \Pi_\varepsilon: \begin{cases} A \subset \Pi_\varepsilon \\ S(\Pi_\varepsilon) < \varepsilon \end{cases}$$



$\uparrow \textcircled{1}$
 $\Rightarrow$

$$0 \leq S(A) \leq S(\Pi_\varepsilon) < \varepsilon \Rightarrow$$

т.к. $\varepsilon - \forall$

$$S(A) = 0$$

$$3) \quad \exists \begin{cases} A \cap B = C \\ S(C) = 0 \end{cases} \Rightarrow S(A) = S(A \setminus C) + S(C) = S(A \setminus C)$$

$$\Rightarrow S(A \cup B) = S(A \setminus C) + S(B) = S(A) + S(B)$$

Вычисление площади плоской фигуры

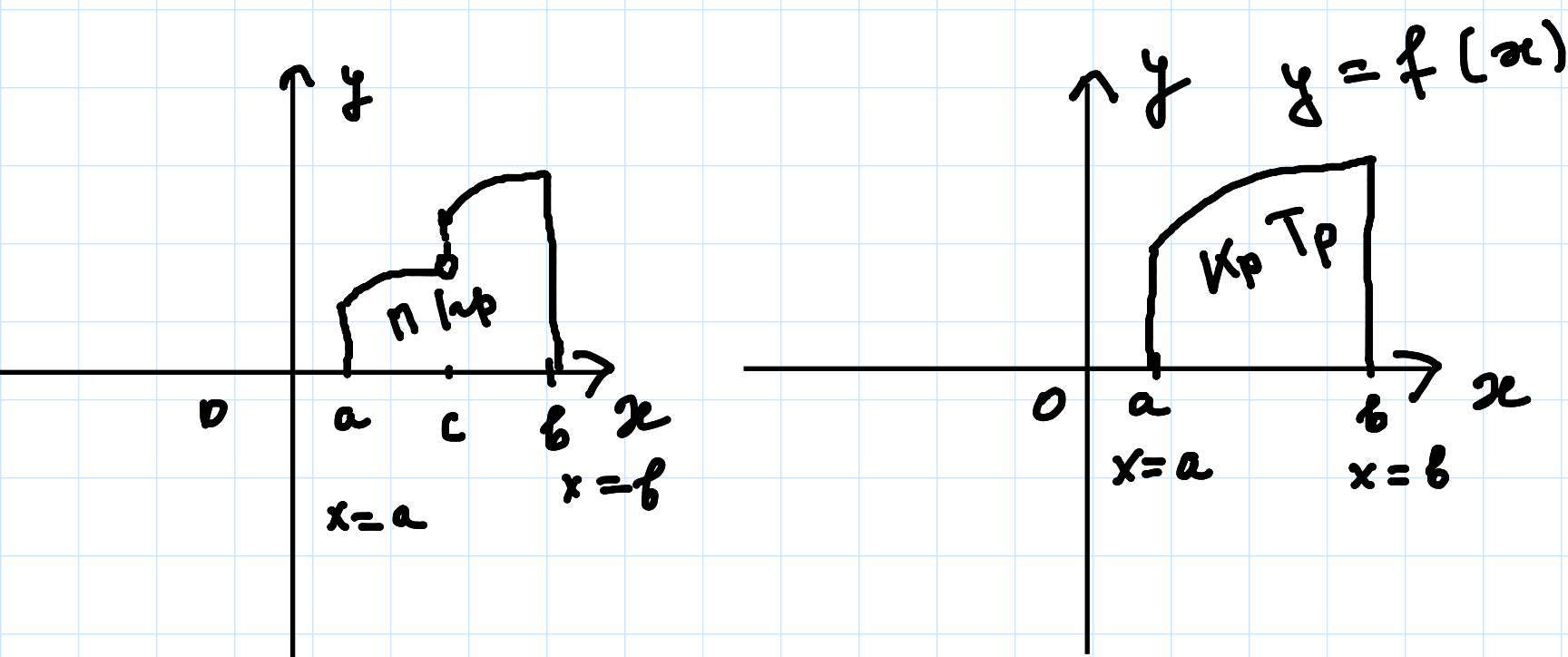
Опр $\exists f: [a, b] \rightarrow \mathbb{R}, \quad \boxed{f \geq 0}$

Множ-во

$$G_f = \left\{ (x, y) \in \mathbb{R}^2 : x \in [a, b], \right. \\ \left. 0 \leq y \leq f(x) \right\}$$

наз. подграфиком ф-ции f .

Если $f \in C^0 [a, b]$, то подграфик ф-ции
наз. криволинейной трапецией.



(Т) $\exists f \in R [a, b], \quad G_f$ - н/р f .

Если $\exists S(G_f)$, то

$$S(G_f) = \int_a^b f(x) dx$$

$\Delta \exists f \in R [a, b], \quad \exists \tau: [a, b]$

$\exists \underline{S}_\tau(f), \quad \bar{S}_\tau(f) \Rightarrow$

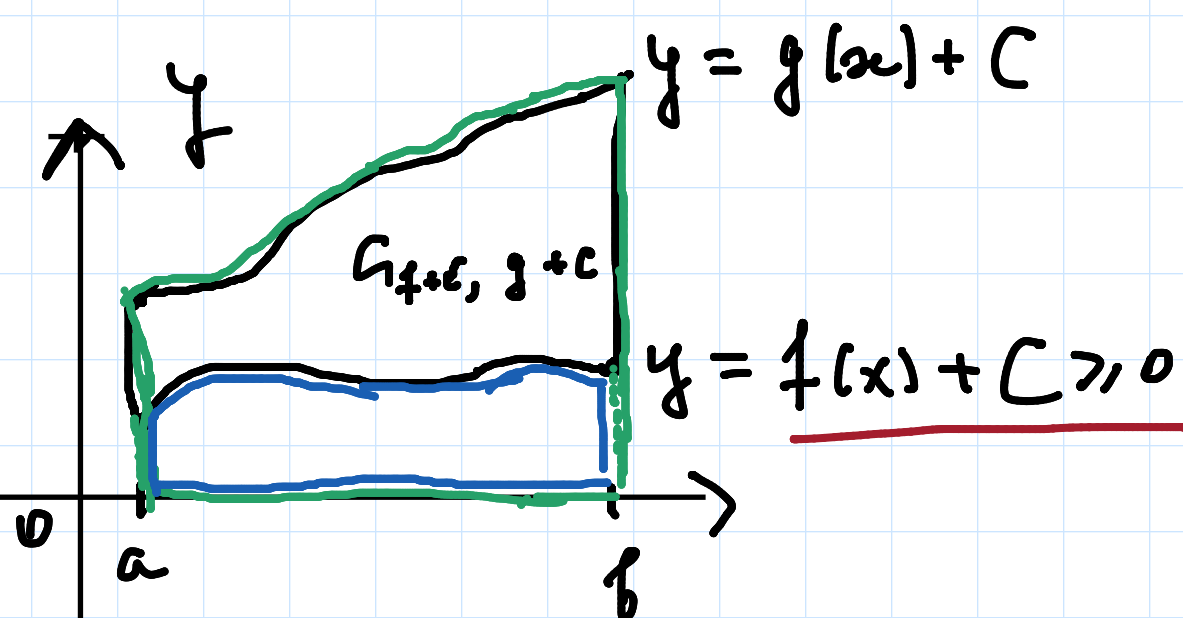
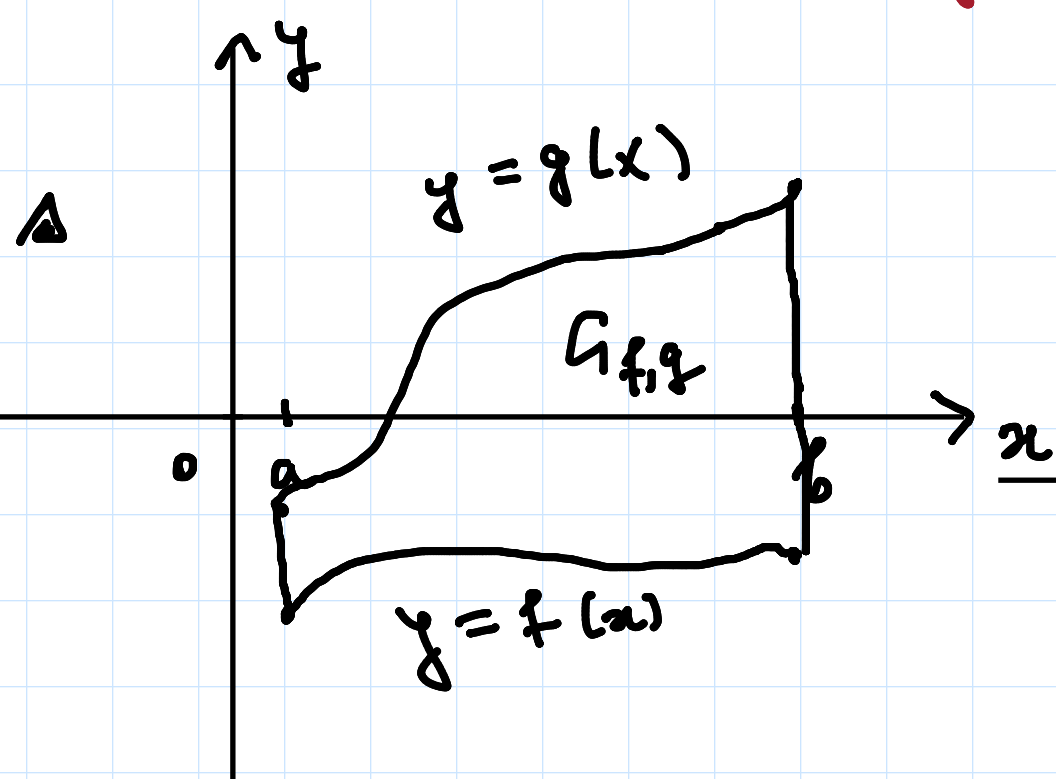
$$\begin{aligned} \underline{S}_\tau(f) &\leq S(G_f) \leq \bar{S}_\tau(f) \\ \downarrow \lambda(\tau) \rightarrow 0 & \quad \quad \quad \downarrow \lambda(\tau) \rightarrow 0 \\ \int_a^b f(x) dx & \quad \quad \quad \int_a^b f(x) dx \end{aligned}$$

$$\rightarrow S(G_f) = \int_a^b f(x) dx \quad \Delta$$

$$\textcircled{T} \quad \exists f, g \in R[a, b], \quad f \leq g$$

$$\exists G_{f,g} = \{(x, y) \in \mathbb{R}^2 : x \in [a, b], f(x) \leq y \leq g(x)\}$$

$$\rightarrow S(G_{f,g}) = \int_a^b (g(x) - f(x)) dx.$$



$$\begin{aligned} S(G_{f,g}) &= S(G_{f+c, g+c}) = S_{\text{gen.}} - S_{\text{cur.}} = \\ &= S(G_{g+c}) - S(G_{f+c}) = \\ &= \int_a^b (g+c) dx - \int_a^b (f+c) dx = \\ &= \int_a^b g dx + c \int_a^b dx - \int_a^b f dx - c \int_a^b dx = \\ &= \int_a^b (g(x) - f(x)) dx. \quad \Delta \end{aligned}$$

Опр.

$$\exists 0 < \beta - \alpha \leq 2\pi$$

$$f : [\alpha, \beta] \rightarrow \mathbb{R} \quad f > 0$$

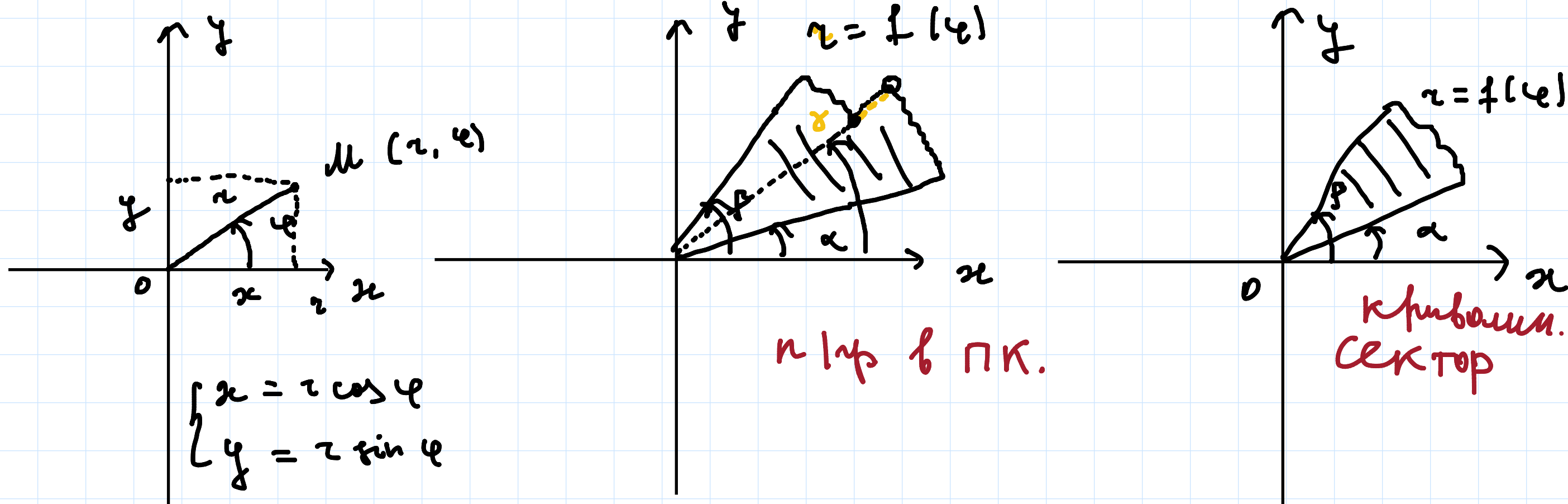
$$\text{Мн во } \tilde{G}_f = \{(r \cos \varphi, r \sin \varphi) \in \mathbb{R}^2 :$$

$$\varphi \in [\alpha, \beta]$$

$$0 \leq r \leq f(\varphi)\}$$

наз. полярными координатами φ -угол f по полярным координатам x .

Если $f \in C^0 [\alpha, \beta]$, то построим
криволинейным секторам.



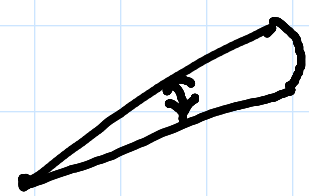
(I) $\exists f \in R [\alpha, \beta]$, $\tilde{G}_f$ — н/ч f
Если $\exists S(\tilde{G}_f)$, то

$$S(\tilde{G}_f) = \frac{1}{2} \int_{\alpha}^{\beta} f^2(\varphi) d\varphi$$

$\Delta \tau = \{\varphi_i\}$, $i=0, n$ — разбиение $[\alpha, \beta]$

$$\Delta \varphi_i = \varphi_i - \varphi_{i-1}, \quad m_i = \inf_{\Delta_i} f(\varphi)$$

$$M_i = \sup_{\Delta_i} f(\varphi)$$



$$S_{\text{сек}} = r^2 \frac{\varphi}{2}$$

$$S_{\tau}(f) = \frac{1}{2} \sum_{i=1}^n m_i^2 \Delta \varphi_i$$

$$S_{\tau}(f) = \frac{1}{2} \sum_{i=1}^n M_i^2 \Delta \varphi_i$$

сумма

Далее покажем $\frac{f^2(\varphi)}{2}$

$$S_{\tau}(f^2/2) \leq S(\tilde{G}_f) \leq S_{\tau}(f^2/2)$$

т.к. $f^2(\varphi) \in R[\alpha, \beta] \Rightarrow$

$$\exists \int \lim_{\lambda|\tau| \rightarrow 0} S_{\tau}(f^2/2) = \frac{1}{2} \int_{\alpha}^{\beta} f^2(\varphi) d\varphi$$

$$\lim_{\lambda|\tau| \rightarrow 0} S_{\tau}(f^2/2) = \frac{1}{2} \int_{\alpha}^{\beta} f^2(\varphi) d\varphi$$

$$\Rightarrow S(\tilde{G}_f) = \frac{1}{2} \int_a^b f^2(x) dx$$